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Faith? Hope? Charity? Religion explains giving when warm glow and impure altruism do not

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Abstract

I explain why many charities, especially religious ones, spend significantly to improve poor community members' living standards. Conventional theoretical philanthropy models with altruism or impure altruism/warm glow cannot explain poverty-alleviating private transfers in the type and size of communities where this occurs—for example, congregations. However, if preferences reflect religious precepts (you are your brothers' keeper; treat neighbours like yourself), charitable poverty alleviation can be optimal even in large communities. Income and price elasticities for giving prove important. Surprisingly, charitable private transfers are less likely when donors care for recipients' living standard instead of the aggregate amount all recipients get.

KEYWORDS

altruism, philanthropy, poverty alleviation, public good, religion, warm glow

JEL CLASSIFICATION

D640, H410, L310, Z120

1 | INTRODUCTION

“What we do not know dwarfs what we do know about the economics of charity” (List, 2011, 177). This motivates studying neglected aspects like charitable poverty alleviation, this paper's focus. Using central religious tenets, I explain why: (i) religion gets so much charity; (ii) proceeds go significantly

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to reduce poverty. I show (Sections 3 and 5.1) that the “Nash contribution” and joint-product/warm-glow models imply poverty-reducing transfers need not raise total welfare in large populations if competing with other charitable uses, such as pure public goods. This result derives from the second of two large-number effects in voluntary provision of a public good and a private good. The first effect causes free-riding and public good undersupply. The second arises as a transfer directly benefits one recipient but its opportunity cost rises with the beneficiaries from a public good that it could have financed.¹ Warm glow overcomes just the first effect. It does not explain giving to reduce poverty in large communities, but preferences embodying central religious tenets do.

Christian tenets supporting benevolence include “Without charity, I am nothing and I gain nothing...” (1 Corinthians 13:1–13). So, in a pious USA, charitable giving is large: cash giving matches clothes spending, half spending on health from post-tax income (Bureau of Labour Statistics, 2013). Religion gets 35%–40% of all U.S. giving, thrice the second-ranked, education (Andreoni & Payne, 2013; List, 2011). In the UK, it is joint-top with health.

My analysis embraces all major faiths. These all have “pro-social” precepts stressing care for the poor, which are analogous to preferences for redistribution.² Judaism's *tzedakah* (righteousness via charity) is an obligation. Islam's third of five pillars is *zakat*—alms. “Dana” (charity) underpins *dharma* (righteousness) in Hinduism, Buddhism, Sikhism and Jainism. Hinduism says money not shared with the poor gives no happiness. The Bhagavadgita (3.19–26) commands “work with others' welfare in mind” and the Tamil Tirukkural says: “Giving to the poor is true charity, all other giving expects some return”.

Much of religious giving reduces poverty. In 1989, congregations got half of total US charity receipts of near \$100bn (Biddle, 1992), spending \$20bn on members' “specific material needs”.³ Allard (2008) shows religious charities focus more on such needs than secular charities and, since 1997, have offset U.S. governments' reduced transfers to the poor.

Economists rather neglect poverty-reduction-motivated charity. They argue that donors' altruism towards, for example, transfers, day-care or housing recipients makes these public goods. They then study the public good, but not the private benefit, aspects of charity. I argue that one must study both aspects for a full understanding of charity. Formally, this is the crucial difference between my work and the existing literature. My analysis here and in Fraser (2006) are the only ones I know that bundle public good and private good provision when there is an opportunity cost of providing one in terms of the other. I believe that models in the literature that are interpreted as involving private transfers to the poor do not involve such transfers. They involve, as Bergstrom et al. (1986) title states accurately, “private provision of public goods”. Unlike in my analysis, they do not have *any* individual increasing

¹Fraser (1996) shows that this second effect could make it socially optimal to make a poor person worse off by forcing them to contribute to an excludable public good. Fraser (2006) seems the first model of competing uses of charitable donations for private transfers or a public good where this effect operates.

²There are many possible origins and formulations of preferences for redistribution, with varying implications. Alesina and Giuliano (2011, p. 96) note that they are “interested specifically in the determinants of preferences for redistribution”. They survey the theoretical models and associated empirical evidence, focussing especially on religion and race. Based on the empirics, they argue (p. 113): “*Being brought up in a religious environment has the effect of increasing tastes for redistribution independently of the religious denomination.*” They study cases where utility depends indirectly or directly on a measure of inequality. I consider preferences for redistribution consistent with being religious specifically to see the implications for giving to the poor. By making utility increasing in transfers to the poor in Section 4 below, I make redistribution more likely in the equilibrium of my model.

³“60% of all congregations offered a food pantry, 52% offered clothing closets, ... 43% had programs for visiting the sick and 41% sponsored a soup kitchen...” Cnaan and Yancey (2000). Comprehensive recent data on all aspects of giving can be found at The Philanthropy Roundtable (e.g., <https://www.philanthropyroundtable.org/almanac/statistics/who-gives>).

their consumption of a private good as a consequence of receiving transfers. So, what is donated goes entirely and directly to providing a public good, and that's all.

Some studies justify neglecting charitable transfers on the grounds that state welfare crowds out poverty-reducing charity (e.g., Roberts (1984)). Few later crowding-out studies examine poverty reduction.⁴ Boberg-Fazlić and Sharp (2017) do, finding “strong empirical support for a positive relationship between welfare provision and charitable giving”—that is, they find crowding-in.

The aspects of religions that seem most relevant for charity are precepts obliging help to the poor. I capture these by including total spending on transfers, separate from that on a pure public good, in all utility functions. This reflects a taste for redistribution as it yields utility from improving the poor's welfare, not just from the act of giving itself or giving directly benefiting oneself. It captures List's (2011, p. 165) idea that different motives to other types of charity spur religious giving. He advances social insurance: “*people give to churches so that in time of need they can more readily receive Church services or help from fellow churchgoers.*” I call such preferences, arguably required by scriptures, “pious”. Section 4 shows that a charity for such agents can find transfers to the poor welfare-enhancing in large populations as the transfers' social utility increases with the population.

Superficially, it might seem more natural to reflect concern for poor brethren's welfare by preferences based on per capita, not total, transfers, as per capita transfers are what determine a poor person's total income, consumption, and hence welfare. However, I do not assume so here for several reasons. First, total transfers can be viewed as an “insurance pool” in List's interpretation. Poor individuals' welfare can be affected both by the per capita transfer to them, and also the existence of that pool. Second, with a fixed population of beneficiaries, a one-one link exists between total and per capita transfers, so an increase in the former increases the latter. Third, I believe that a redistributive charity like the Salvation Army and its constituency focuses more on what it does collectively for all rather than for each individual.

More concretely, I show in Section 5.2 (supported by the Online Appendices) that tastes based on per capita transfers [which I call “Brown preferences” after Brown (1997)] “do not work” as well as pious preferences to create an environment in which transfers to the poor can be optimal in large congregations when such transfers compete with a public good. First, other things equal, Brown preferences generate a feasible subgame-perfect Nash equilibrium in charitable giving in very small congregations. However, with typically-sized congregations, they are much less likely to generate such an equilibrium than when agents have pious preferences. Second, other things equal, for a given fraction of donations devoted to alleviating poverty by the charity, less is likely to be given with tastes based on per capita transfers than on total transfers. Lastly, considering the impact on giving of the proportion of the congregation of fixed size that is poor, I show that we get diametrically opposite results with Brown and pious preferences. With the former, the amount given to charity by each donor and the share devoted to transfers decreases as the proportion of the poor in the congregation increases. With pious preferences, the opposite is true.

My model augments three empirical literature strands. One (Clark & Lelkes, 2006; Scheve & Stasavage, 2006) links faith to redistribution. Faith insures against mishaps by giving psychic benefits as, say, health, jobs, or pensions. A welfare state is smaller as these psychic benefits replace state ones. Another strand (Hungerman, 2005) says that Church welfare substitutes for the state's. Again, a welfare state is smaller in more pious societies. A third says religious giving insures for an afterlife (Chang, 2005; Hrung, 2004; Thornton and Helms, 2013).

⁴de Wit and Bekkers' (2017) extensive meta-analysis concludes that “private giving is generally not crowded out by public funding”. They focus on crowding out in public good supply and ignore transfers.

My analysis predicts that, if tastes reflect religion's precept of sharing with the poor, charitable transfers occur. This rationalises the first two empirical strands. But, it is not just the pious who give to the poor; the irreligious might also give if they have prosocial tastes that value transfers to the poor. My analysis does not preclude this. Nevertheless, like Alesina and Giuliano (2011), List (2011), and others, I recognize that religion is sufficiently important in the charity landscape to justify special attention. The Conclusion studies the afterlife.

I show that giving part of charitable donations to the poor can increase giving even if donors get no altruistic externality from such transfers. An altruistic externality from transfers can raise giving and welfare, explaining transfers. Section 2 reviews models without religion; Section 3 shows the difficulty, with “standard” tastes, of reducing poverty via transfers alongside public good supply. Section 4 shows pious tastes overcome this theoretically and numerically. Section 5 examines other tastes, including joint-product/warm-glow ones. Section 6 concludes. An Appendix has some proofs; Online Appendices contain computations, more proofs and discussion.

2 | RELATED LITERATURE: GIVING WITHOUT RELIGION

Key papers I build on are: Becker (1974), Warr (1982, 1983), Cornes and Sandler (1984, 1994), Roberts (1984), Bergstrom, Blume and Varian (BBV) (1986), Andreoni (1988, 1989, 1990), Fraser (1992, 2006), Vicary (1997), Ribar and Wilhelm (2002) and Faias et al. (2014). All, bar Becker, Warr (1982), Roberts, and Fraser (2006), study charitable public good supply without transfers.⁵

BBV's Nash equilibrium model gets striking findings, including Warr's (1983) neutrality (redistribution among donors so that none loses more than their original donation leaves total giving unchanged), via Assumption 1:

Assumption 1. Individuals (i 's) have quasi-concave utility functions $u^i[x^i, G]$, $i = 1, \dots, n$, defined over private good consumption, x^i , and public good consumption, G , which are non-decreasing in their arguments. Incomes (m^i) are exogenous. $G = \sum_i g^i$ sums individual contributions ($g^i \geq 0$) to providing the public good.

Andreoni (1988) extends BBV for a limit result: in large economies hardly anyone gives.

Economists agree that this “Nash model” predicts “too little” giving. Cornes and Sandler's (1984) and Andreoni's (1989, 1990) “joint product”/“warm-glow” models address this issue. In their models, donors benefit from the public good and giving per se. Still using u for a utility function, theirs is $u^i[x^i, g^i, G]$. The g^i , a private good for i , gives i an extra spur to donate, altering the neutrality and limit results (cf. Andreoni, 1989, 1990; Ribar & Wilhelm, 2002).^{6,7}

I modify Assumption 1, relabelled Assumption 3, to study charitable transfers and public good supply in Section 3. In Section 4, with pious preferences, everyone gets utility from: (i) total transfers

⁵A large literature develops Becker (1974) to study private transfers in households (for example, Cox (1987); Cox and Jakubson (1995) and Cox et al. (2004). However, like de Wit and Bekkers, I only examine charity where donors have no familial links to recipients.

⁶Allouch (2013) elaborates competitive equilibrium with warm-glow. Evren and Minardi (2017) axiomatise warm-glow preferences.

⁷Note that religion cannot be identified with warm glow as the warm-glow model does not embody a preference for redistribution that is integral to all major religions. Warm glow emphasizes the selfishness of giving: you give because you get pleasure from the act of giving itself, irrespective of whether or not that does good to others. As Atkinson (2009, p. 648) notes: “the pure warm-glow model assumes that the donor is completely unconcerned with the use made of the gift and with the effectiveness of charitable activity.”

to the poor; (ii) spending on a shared good. So, it has two crowding-out-prone public goods. Yet it can yield giving and the share used for transfers that are “realistic” in a large society. This relates to Ghosh, Karaivanov and Oak's (GKO, 2007) and Cornes and Itaya's (2010) multiple public goods models. However, they ignore transfers. GKO's two-person model also ignores my population effects while Cornes and Itaya focus on neutrality.⁸

Becker (1974), Warr (1982), Roberts (1984) and Brown (1997) are among few who model charitable transfers. Becker, Warr and Roberts model situations with a sole poor recipient and so do not analyse population issues that are central in my analysis. Brown ignores recipients, apart from the fact that the average donation to them enters her donors' utility functions as an externality. Section 5.2 shows that donors caring for average rather than total donations works against transfers. These authors ignore that charities only make transfers if this enhances total welfare, which depends on the number of consumers of any public good that could be funded instead.⁹

As I focus on a charity's redistribution via direct transfers, my analysis differs from Uler (2009), who does not study this aspect of giving. She considers how redistribution effected by an income tax system impacts giving to charity solely for providing a public good, so the new large numbers issue I consider is not relevant. Uler's tax system entails a flat rate tax on income net of charitable donations (which implies preferential tax treatment of donations) with proceeds returned to all households equally. She establishes, among other things, conditions under which voluntary giving increases in the level of redistribution. Although churches as charities, as well as individual donors, receive preferential tax treatment in reality, that treatment does not usually discriminate between a Church's uses of funds and, thereby, directly influence the issues I focus on. Thus, for brevity, I do not consider it explicitly in my formal analysis.

Maybe closest to my analysis, Kotchen (2006) models a scenario relevant to charity for a good with public and private features. For example, buying a grocery item might yield a fixed donation to a public good. His model's principal differences to mine are: first, its mix of characteristics is fixed exogenously in a one-stage game. In mine, taking the Church/charity as bundling a public good and private transfers, their combination arises endogenously from a strategic interaction between charity and consumers in a two-stage game. Second, while he studies how group size affects equilibrium provision of the public characteristic, his model is not set up to show how group size impacts whether or not charitable transfers occur.

Despite a dearth of theory on charitable transfers or religious charities, an impressive empirical literature exists. Bекkers and Wiepking (2011) review key articles.

3 | POVERTY ALLEVIATION WITHOUT PIOUS PREFERENCES¹⁰

Imagine a congregation run by “a welfarist charity” agent—for example, Church Elders. It knows churchgoers' incomes and identical tastes. In stage 1 of a 2-stage game with them, it collects funds

⁸Faías et al. (2014) extend BBV's results to multiple public and private goods.

⁹Three recent theoretical analyses with donations used *only* for private transfers are Atkinson (2009), for overseas aid, Echazu and Nocetti (2015), for a discrete good, and Bourlès et al. (2017) for altruism in a network. Both Atkinson and Echazu and Nocetti consider modelling strategies to bypass the free-riding problem with a large number of donors that arises from treating the donations as a public good, which is usually the bane of theoretical models. However, neither Atkinson nor Echazu and Nocetti address the separate large number issue arising specifically from the competing uses for donations that is my focus.

¹⁰The basic model here mirrors that in early versions of Fraser (2006). That paper notes, via a numerical example, the difficulty of making private transfers optimal in large populations. It neither analyses tastes overcoming this (“pious” preferences here), nor others that also fail to make private transfers optimal.

for two mutually-exclusive stage-2 uses: (i) public goods like upkeep of the Church and pastor; (ii) poverty-reducing private goods like food, clothing and shelter. It creates an altruistic externality and public good if, while not consuming the goods (e.g., soup kitchens), donors get utility as recipients benefit. It also gives private goods to soup kitchen patrons. I study cases without the externality here, and with it in Section 4. The charity fixes the shares of total donations for (i) and (ii), mindful how they decide giving. Even if donors do not care for reducing poverty, the charity does via its overall welfare concern.

To focus sharply on poverty alleviation, my “two-class model” assumes:

Assumption 2. There are two types, indexed h and l , respectively, with exogenous “high” and “low” incomes $m^i > 0$, $i = h, l$; $m^h > m^l$. There are n^i ($n^i \geq 1$) individuals of type i , $i = h, l$.

Assumption 3. Respectively, x and G are amounts of infinitely divisible sole private and public goods of unity unit costs. Both types have a concave increasing utility function $U(x; G)$ (with subscripts for partial derivative, $U_i > 0, U_{ii} \leq 0$, $i = x, G$; $U(x, G)$ has a negative semi-definite Hessian). U denotes utility function throughout, whatever the argument vector.

Assumption 4. (i) The utility function is supermodular: $\partial^2 U / \partial x \partial G \equiv U_{xG} \geq 0$. (ii) Each good is essential and satisfies an Inada condition: $\lim_{x \downarrow 0} U_x(x, G) = \infty$, $\lim_{G \downarrow 0} U_G(x, G) = \infty$.

Essentiality means that consumers with resources consume both goods. Supermodularity ensures monotone comparative statics. It implies complementarity between the goods.

A richer model of many income types could endogenize which type(s) qualify for transfers. For tractability, I skirt this. Tractability also defends Assumption 3. Better defences are: (i) taste differences exist, but economists rather explain behaviour differences by prices and incomes; (ii) congregations might have similar tastes à la Tiebout. With taste differences, well-being is not measured just by income. But, practically, charities separate “rich” from “poor” by income. Consumers satisfying (Assumption 3) get no altruistic externality from transfers to the poor.

3.1 | The charity

At stage 2, the utilitarian charity,¹¹ sole provider of the public good, uses donations for it and for transfers to the l types in shares fixed at stage 1 reflecting concern for inequality:

Assumption 5. The charity uses a share $\lambda \in [0, 1]$ of receipts for the public good and $1 - \lambda$ for transfers. It can identify the poor and gives each $(1 - \lambda)/n^l$ of a transfer of $(1 - \lambda)$.

Assumption 5 implies that, as λ decreases, it costs more for an h type to augment G , and less for an l type if h types also give.

3.2 | Consumer behaviour and the stage-2 Nash equilibrium

For the charity's chosen λ , an arbitrary l type conjectures that the $n^l - 1$ others each donates g^l and the n^h h types give g^h that it cannot affect. If it gives g^l , it supposes it gets a transfer, c^l , satisfying $n^l c^l = (1 - \lambda) \left[n^h \bar{g}^h + (n^l - 1) \bar{g}^l + g^l \right] = (1 - \lambda) (G + n^l c^l)$, that is,

¹¹Bilodeau and Slivinski (1997) study different charity motivations.

$$c^l = \left(\frac{1-\lambda}{n^l} \right) \left[n^h \bar{g}^h + (n^l - 1) \bar{g}^l + g^l \right] = \left(\frac{1-\lambda}{\lambda n^l} \right) G \quad (1)$$

$$x^l = m^l + c^l - g^l = m^l + \left(\frac{1-\lambda}{n^l} \right) \left[n^h \bar{g}^h + (n^l - 1) \bar{g}^l \right] - \left(\frac{n^l + \lambda - 1}{n^l} \right) g^l \quad (2)$$

Let $G_{-l} \equiv \lambda \left[n^h \bar{g}^h + (n^l - 1) \bar{g}^l \right]$ be others' total donations to public good supply. By

$$G = G_{-l} + \lambda g^l \Rightarrow g^l = \left(\frac{G - G_{-l}}{\lambda} \right) \quad (3)$$

and Equations (1) and (2), the arbitrary l -type's rewritten budget constraint is

$$m^l + \frac{G_{-l}}{\lambda} = x^l + \left[\frac{n^l + \lambda - 1}{\lambda n^l} \right] G \quad (4)$$

The LHS Equation (4) is a type l 's “full income”. Their effective price for G is $(n^l + \lambda - 1) / \lambda n^l$. Defining G_{-h} similarly and using $x^h = m^h - g^h$, an h type's budget constraint is

$$m^h + \frac{G_{-h}}{\lambda} = x^h + \left[\frac{G}{\lambda} \right] \quad (5)$$

Substituting Equations (4) and (5) into the types' utility functions give their maximisation problems:

$$\text{Max.}_{G \geq G_{-h}} U \left[m^h + \frac{G_{-h}}{\lambda} - \frac{G}{\lambda}, G \right] \quad (6)$$

$$\text{Max.}_{G \geq G_{-l}} U \left[m^l + \frac{G_{-l}}{\lambda} - \left(\frac{n^l + \lambda - 1}{\lambda n^l} \right) G, G \right] \quad (7)$$

I study a subgame perfect equilibrium. Stage 2 involves a Nash equilibrium. With g^i , $i = h, l$, as choice variables, this is a non-negative donations vector, $\{g^h, g^l\}$, such that:

(a) for each type h , g^h solves

$$\begin{aligned} & \text{Max.}_{g^h \geq 0, x^h \geq 0} U [x^h, G_{-h} + \lambda g^h] \\ & \text{s.t. } m^h = x^h + g^h \end{aligned} \quad (8)$$

(b) for each type l , g^l solves

$$\begin{aligned} & \text{Max.}_{g^l \geq 0, x^l \geq 0} U [x^l, G_{-l} + \lambda g^l] \\ & \text{s.t. } m^l = x^l - \left(\frac{1-\lambda}{\lambda n^l} \right) G_{-l} + \left(\frac{n^l + \lambda - 1}{n^l} \right) g^l \end{aligned} \quad (9)$$

I only study equilibria wherein all in a type behave identically. Abusing terminology, I call these symmetric equilibria. For given λ , Fraser (2006, Theorem 1), after Andreoni and Bergstrom (1996), shows a stage-2 equilibrium exists uniquely. I use the following shorthand:

$$L(g^h, g^l, \lambda) \equiv U \left[m^l + \left(\frac{1-\lambda}{\lambda n^l} \right) G_{-l} + \left(\frac{1-\lambda-n^l}{n^l} \right) g^l, G_{-l} + \lambda g^l \right]; \quad (10)$$

$$H(g^h, g^l, \lambda) \equiv U \left[m^h - g^h, G_{-h} + \lambda g^h \right] \quad (11)$$

With the same $U(\cdot)$, if I omit utility arguments, superscripts in U^i , $i = h, l$, separate types. With all taking others' donations as given and subscripts showing partial derivatives, first-order conditions (FOCs) for a Nash equilibrium are the Kuhn–Tucker conditions

$$L_{g^l} = \left(\frac{1-\lambda-n^l}{n^l} \right) U_x^l + \lambda U_G^l \leq 0, \quad g^l \geq 0 \quad (12)$$

$$H_{g^h} = -U_x^h + \lambda U_G^h \leq 0, \quad g^h \geq 0 \quad (13)$$

By K–T, the inequalities in each of systems (12) and (13) are not both strict. By essentiality, $g^h > 0$ and/or $g^l > 0$, so L_{g^l} and/or H_{g^h} equals zero. By the implicit function theorem, if types' Jacobians are invertible at the optimum, non-zero donations solving Equations (12) and (13) are locally differentiable functions in the parameters. I suppress parameters like m^i and n^i and focus only on λ , writing optimal donations as Nash reaction functions, $g^i(\lambda)$, $i = l, h$.

Problems (6) and (7) (equivalently, Problems (8) and (9)) are analogues to BBV's (1986). They are standard, bar constraints $G \geq G_{-i}$, $i = h, l$, stopping choice of G below what others jointly finance. With Assumption 3 and Assumption 4 (ii), demand theory implies Problems (6) and (7) yield public good demands $f[m^i + (G_{-i}/\lambda), p_G^i]$, $i = l, h$, for a function $f(\cdot)$ of the respective type's full income and public good price (respectively, $p_G^h = 1/\lambda$; $p_G^l = (n^l + \lambda - 1/\lambda n^l)$). These satisfy $G \geq f[m^i + (G_{-i}/\lambda), p_G^i]$, $i = l, h$, as no-one's optimal G can exceed its supply. By essentiality, $G > 0$ and at least one of these last weak inequalities is an equality. If the two types with identical utility functions donate, they do not equalize private consumption and utilities, unlike in BBV. If $\lambda < 1$, an l has a lower price for G , yet demands the same as an h . So, with G normal, an l type has less equilibrium income and private consumption. They could donate despite getting transfers as they find giving relatively cheap, but their incentive to do so falls fast as n^l increases.

3.3 | The charity's maximisation

At stage 1, the charity optimises subject to consumers' reaction functions, choosing $\lambda \in [0, 1]$ to maximise utilitarian social welfare (summed utilities),

$$W(\lambda) \equiv n^h H[g^h(\lambda), g^l(\lambda), \lambda] + n^l L[g^h(\lambda), g^l(\lambda), \lambda] \quad (14)$$

The overall equilibrium concept is subgame perfection. As the charity knows congregants' incomes and utility function, so their stage-2 Nash equilibrium choices, g^i , $i = h, l$, in expression (14), its stage-1 choice λ decides the entire equilibrium. By the continuous differentiability of U , and the implicit function theorem, derived utilities H and L are continuously differentiable in λ , thus so is W . Then, by Weierstrass' extreme value theorem, W attains maxima and minima w.r.t. λ on the compact interval

[0; 1]. But, as W 's concavity cannot generally be established, I cannot say an optimum is unique. So, I proceed piecemeal.

By essentiality, no charity with resources opts for $\lambda = 0$ (i.e., $G = 0$). However, income distributions exist with both types donating at $\lambda = 1$ and no transfers. Incomes are then equalized in Nash equilibrium and $\frac{\partial W}{\partial \lambda} \Big|_{\lambda=1} \geq 0$ (Fraser, 2006). Fraser (2006) also shows that the poor never donate in utilitarian equilibria with transfers. So, for distributions with only rich donors when $\lambda = 1$, I explore when transfers occur, which must be welfare-improving. Then two cases exist: (a) there is a W -maximising $\lambda < 1$ and $\frac{\partial W}{\partial \lambda} \Big|_{\lambda=1} < 0$; if W decreases in λ at $\lambda = 1$, reducing λ from $\lambda = 1$ increases W for some $\lambda \in (0, 1)$; (b) W is non-decreasing in λ at $\lambda = 1$, so $\frac{\partial W}{\partial \lambda} \Big|_{\lambda=1} \geq 0$, but the W -maximising $\lambda < 1$. If (b) holds, then $(\partial W / \partial \lambda) < 0$ for some $\lambda < 1$. $(\partial W / \partial \lambda) > 0$ for all $\lambda \in [0, 1]$ when $g^h(\lambda) > 0 = g^l(\lambda)$ rules out optimal transfers.

Let $g^i(\lambda)$, $i = h, l$, be derivatives w.r.t. λ of donations at arbitrary λ and $g^l(\lambda) = 0 = g^l'(\lambda)$ at an optimum of $W(\lambda) \equiv n^h U[m^h - g^h(\lambda), \lambda n^h g^h(\lambda)] + n^l U[m^l + ((1 - \lambda)/n^l) n^h g^h(\lambda), \lambda n^h g^h(\lambda)]$. Differentiating (with superscripts again showing types without functional arguments),

$$\frac{\partial W}{\partial \lambda} = \underbrace{[n^h U_G^h + n^l U_G^l - U_x^l] n^h g^h}_{\text{an "income effect"}} + \underbrace{[(n^h - 1) \lambda U_G^h + \lambda n^l U_G^l + (1 - \lambda) U_x^l] n^h g^{h'}}_{\text{a "price effect"}} \quad (15)$$

using $U_x^h = \lambda U_G^h$ from Equation (13). Also, differentiating $H_{g^h} = 0$, $g^{h'} = -[U_{xx}^h - 2\lambda n^h U_{Gx}^h + \lambda^2 n^h U_{GG}^h]^{-1} [\lambda n^h g^h U_{GG}^h + U_G^h - n^h g^h U_{Gx}^h]$. So, $U_{Gx}^h \geq 0$ and $-GU_{GG}^h/U_G^h \geq 1$, with one inequality strict, suffice for $g^{h'}(\lambda) < 0$; an additively separable $U(\cdot)$ and $-GU_{GG}^h/U_G^h < 1$ give $g^{h'}(\lambda) > 0$. We easily find cases of $g^l(\lambda) = 0 = g^l'(\lambda)$ and $g^{h'}(\lambda) < 0$, so, increasing transfers (reducing $\lambda \leq 1$) increases total giving.¹² Is this welfare-improving in reasonably-sized communities like congregations? This needs $\partial W / \partial \lambda < 0$.¹³

Equation (15) shows how community size determines if transfers are desirable. The “income effect” is the opportunity cost of reducing λ (utility foregone from G), using funds released for transfers. Reducing λ also changes giving by $n^h g^{h'}(1)$. The “price effect” is the welfare effect of this altered giving, net of its impact on donors' private consumption.

Note $g^{h'}(\lambda) < 0$ is unnecessary nor suffices for $\partial W / \partial \lambda < 0$. If $g^{h'}(\lambda) \geq 0$, $\partial W / \partial \lambda < 0$ can hold if $[n^h U_G^h + n^l U_G^l - U_x^l] n^h g^h < 0$. But, we see readily, if $g^h(\lambda) > 0 = g^l(\lambda)$, this is unlikely for large n^h and/or large n^l , as Proposition 1 (proven in the Appendix) shows.

Proposition 1. *If $g^h(\lambda) > 0 = g^l(\lambda)$, in Equation (15), $[n^h U_G^h + n^l U_G^l - U_x^l] n^h g^h > 0$ at $\lambda \leq 1$ if n^h and/or n^l is sufficiently large.*

With U specified, we can verify if $[n^h U_G^h + n^l U_G^l - U_x^l] n^h g^h > 0$ for an actual (n^h, n^l) .

¹²This result that more redistribution can increase giving was in Fraser (2006) and even earlier versions. As I show presently, it is also easy to find examples where $g^{h'} < 0$ does not hold.

¹³Andreoni and Payne (2013) caution against inferring anything increasing donations is good for society.

Next, to closely study what determines $g^{h'}(\lambda)$, I use properties of the public good demand function, $G \equiv f \left[m^h + \frac{G-h}{\lambda}, \frac{1}{\lambda} \right] = \lambda n^h g^h(\lambda)$. Proposition 2, proven in the Appendix S1, relates $\text{sign} \left(g^{h'}(\lambda) \right)$ to the price and income elasticities of giving.

Proposition 2. Let f_m and f_p , respectively, be derivatives of $f(\cdot)$ w.r.t. an h 's full income, $m \equiv m^h + (n^h - 1) g^h(\lambda)$ in a symmetric equilibrium, and their public good's effective price, $p \equiv 1/\lambda$. Let $\varepsilon_m \equiv m f_m / f$ and $\varepsilon_p \equiv \lambda^{-1} f_p / f$, respectively, be the income and price elasticity of demand for the public good. For “large” n^h , $\varepsilon_p > -1$ and $\varepsilon_m \leq 1$ suffice for $g^{h'}(\lambda) < 0$.

Using $g^h(\lambda) = f \left[m^h + (n^h - 1) g^h(\lambda), \frac{1}{\lambda} \right] / \lambda n^h$, Proposition 2's proof shows that

$$g^{h'}(\lambda) \cong -(\lambda^2 n^h)^{-1} \left[(1 + \varepsilon_p) / (f^{-1} - \lambda^{-1} \varepsilon_m m^{-1}) \right] \quad (16)$$

No consensus exists on ε_p or ε_m . Taussig (1967) finds small ε_p , -0.04 to -0.10 , and ε_m significantly above 1.0 . Auten et al. (2002) find larger ε_p but smaller ε_m : Their favoured estimates for transitory ε_p vary from -0.40 to -0.61 and persistent ε_p from -0.79 to -1.26 . They get persistent ε_m of 0.40 – 0.87 and transitory ε_m of 0.29 – 0.45 . Karlan and List (2007) find $0 \geq \varepsilon_p \geq -1$ in a natural experiment. Feldstein (1975) and others (e.g., Scharf & Smith, 2015) disaggregate to get $|\varepsilon_p| < 1$ for religion but $|\varepsilon_p| > 1$ for education, hospitals, health and welfare. He finds $\varepsilon_m > 1$ for hospitals and educational entities but $\varepsilon_m < 1$ for others. Such findings imply possibly different empirical signs for $g^{h'}(\lambda)$ for different types of charity. A trend for $0 < \varepsilon_m < 0$ and $-1 < \varepsilon_p < 0$ for religious giving implies $g^{h'}(\lambda) < 0$ might hold if either my model here or in Section 4 applies.

3.4 | The Cobb–Douglas example

A symmetric Nash equilibrium with only h persons giving satisfies FOCs

$$\lambda U_G \left[m^l + \left(\frac{1-\lambda}{n^l} \right) n^h g^h(\lambda), \lambda n^h g^h(\lambda) \right] \leq \left(\frac{n^l + \lambda - 1}{n^l} \right) U_x \left[m^l + \left(\frac{1-\lambda}{n^l} \right) n^h g^h(\lambda), \lambda n^h g^h(\lambda) \right] \quad (17a)$$

$$\lambda U_G [m^h - g^h(\lambda), \lambda n^h g^h(\lambda)] = U_x [m^h - g^h(\lambda), \lambda n^h g^h(\lambda)] \quad (17b)$$

Assume a log-Cobb–Douglas $U(x, G) = \alpha_1 \log x + \alpha_2 \log G$, $\alpha_i \in (0, 1)$, $i = 1, 2$ (cf. Ribar & Wilhelm, 2002, and Andreoni, 2006, without warm glow). Then Equation (17b) implies $\alpha_1 / (m^h - g^h(\lambda)) = \alpha_2 / n^h g^h(\lambda)$, or $g^h(\lambda) = \alpha_2 m^h / (\alpha_2 + \alpha_1 n^h)$, independent of λ . So $g^{h'}(\lambda) = 0$ now.¹⁴ Simplifying, Equation (17a) implies

$$\frac{m^h}{m^l} \geq \frac{n^l (\alpha_2 + \alpha_1 n^h)}{n^h [\alpha_1 n^l - (1 - \lambda) (\alpha_1 + \alpha_2)]} \quad (18)$$

The right hand side (RHS) of inequality (18) is largest with λ close to 0. Even then, as $\alpha_i \in (0, 1)$, $i = 1, 2$, the RHS is approximately $1 + (\alpha_2 / \alpha_1 n^h)$, close to 1 for large n^h . So, inequality (18)

¹⁴ $g^{h'}(\lambda) = 0$ is consistent with $g^{h'}(\lambda)$ in Equation (16). With Cobb–Douglas utility, $\varepsilon_p = -1$, so $g^{h'}(\lambda) = 0$ in Equation (16).

reflects that, in a big group, income inequality need not be great for only the highest income type to give.

From Equation (15), $\text{sign} \left(\frac{\partial W}{\partial \lambda} \Big|_{\lambda \leq 1} \right) = \text{sign} \left[n^h U_G^h + n^l U_G^l - U_x^l \right] \Big|_{\lambda \leq 1}$. As $g^h(\lambda) = \frac{\alpha_2 m^h}{(\alpha_2 + \alpha_1 n^h)}$,

$$\begin{aligned} [n^h U_G^h + n^l U_G^l - U_x^l] \Big|_{\lambda \leq 1} &= (n^h + n^l) \alpha_2 \left[\frac{\alpha_2 + \alpha_1 n^h}{\lambda \alpha_2 m^h n^h} \right] - \frac{\alpha_1 n^l (\alpha_2 + \alpha_1 n^h)}{m^l n^l (\alpha_2 + \alpha_1 n^h) + (1 - \lambda) \alpha_2 m^h n^h} \\ &= \frac{(\alpha_2 + \alpha_1 n^h) [(n^h + n^l) \{ (m^l n^l) (\alpha_2 + \alpha_1 n^h) + (1 - \lambda) \alpha_2 m^h n^h \} - \alpha_1 \lambda n^l m^h n^h]}{\lambda m^h n^h [m^l n^l (\alpha_2 + \alpha_1 n^h) + (1 - \lambda) \alpha_2 m^h n^h]} \begin{cases} > \\ = \\ < \end{cases} 0 \\ \text{as } [(n^h + n^l) \{ (m^l n^l) (\alpha_2 + \alpha_1 n^h) + (1 - \lambda) \alpha_2 m^h n^h \} - \alpha_1 \lambda n^l m^h n^h] &\begin{cases} > \\ = \\ < \end{cases} 0 \end{aligned}$$

In this $(n^h + n^l) m^l n^l \alpha_1 n^h - \alpha_1 \lambda n^l m^h n^h > 0$ if $(n^h + n^l) m^l \geq m^h \geq \lambda m^h$. This is one of alternative sufficient conditions for $\frac{\partial W}{\partial \lambda} \Big|_{\lambda \leq 1} > 0$. For an U.S. congregation of mean size 186 or median size 75, $(n^h + n^l) > m^h / m^l$: median or mean size exceeds the ratio of the incomes of a typical giver and transfer recipient. So, I expect $\frac{\partial W}{\partial \lambda} \Big|_{\lambda \leq 1} > 0$ to hold and no transfers to occur.

4 | POVERTY ALLEVIATION WITH PIOUS PREFERENCES

Assume now everyone cares about total transfers to the poor, T , and tastes satisfy Assumption 6:

Assumption 6. All households have the utility function $U[x, T, G]$, x and G defined as before. U is concave increasing in its arguments, satisfying Inada conditions for x and G .

T does not satisfy an Inada condition as I do not want to make transfers imperative. The private and public goods are indispensable. But, giving to the poor need not be.

A charity behaving as analysed above in a symmetric equilibrium transfers in total

$$T = (1 - \lambda) \left[\sum_{j=1}^{n^h} g_j^h + \sum_{k=1}^{n^l} g_k^l \right] = (1 - \lambda) [n^h g^h + n^l g^l] \quad (19)$$

As $G = \lambda(T + G)$, $T = (1 - \lambda)(T + G) = ((1 - \lambda)/\lambda) G$, at stage 2 a poor person maximises $U(x^l, T, G) = U[x^l, ((1 - \lambda)/\lambda) G, G]$. Their budget constraint in “full income” form (see the Appendix S1) is $x^l + [(n^l + \lambda - 1)/n^l] (G/\lambda) \leq m^l + G_{-l}/\lambda$. So, by non-satiation, their optimisation is

$$\begin{aligned} \text{Max.}_{x^l \geq 0, G \geq G_{-l}} U \left[x^l, \left(\frac{1 - \lambda}{\lambda} \right) G, G \right] \quad \text{s.t.} \quad x^l + \left(\frac{n^l + \lambda - 1}{n^l} \right) \frac{G}{\lambda} &= m^l + \frac{G_{-l}}{\lambda} \\ \Leftrightarrow \text{Max.}_{G \geq G_{-l}} U \left[m^l + \frac{G_{-l}}{\lambda} - \left(\frac{n^l + \lambda - 1}{n^l} \right) \frac{G}{\lambda}, \left(\frac{1 - \lambda}{\lambda} \right) G, G \right] &\quad (20) \end{aligned}$$

Analogously, the arbitrary rich person solves

$$\begin{aligned} & \text{Max.}_{x^h \geq 0, G \geq G_{-h}} U \left[x^h, \left(\frac{1-\lambda}{\lambda} \right) G, G \right] \text{ s.t. } x^h + \frac{G}{\lambda} = m^h + \frac{G_{-h}}{\lambda} \\ & \Leftrightarrow \text{Max.}_{G \geq G_{-h}} U \left[m^h + \frac{G_{-h}}{\lambda} - \frac{G}{\lambda}, \left(\frac{1-\lambda}{\lambda} \right) G, G \right] \end{aligned} \quad (21)$$

The crucial change from non-pious preferences is the new argument in U , $T = (1 - \lambda)G/\lambda$. Invoking Section 3's continuity and differentiability assumptions, the optimisation problems (20) and (21) yield locally differentiable public good demands, $G^p \geq f^{pi} [m^i + (G_{-i}/\lambda), ((1 - \lambda)/\lambda)G_{-i}, p_G^i]$, $i = h, l$; $(p_G^h, p_G^l) = [1/\lambda, (n^l + \lambda - 1)/\lambda n^l]$, with superscript p denoting "from pious preferences".

Again, as only the rich give in an utilitarian charity equilibrium, I examine just $g^{ph}(\lambda) = f^{ph} [m^h + (G_{-h}/\lambda), ((1 - \lambda)/\lambda)G_{-h}, p_G^h] / \lambda n^h$. Differentiating w.r.t. λ , rearranging, using notation as above, plus $\varepsilon_{rel} \equiv \left[\left(T_{-h} f_{T_{-h}}^{ph} \right) / f^{ph} \right]$, the elasticity of giving from concern about others' welfare, yields

$$g^{ph'}(\lambda) \cong -(\lambda^2 n^h)^{-1} \left[\left(1 + \varepsilon_p^p + f_{rel}^{ph} \right) / \left(\frac{1}{f^{ph}} - \lambda^{-1} \frac{\varepsilon_m^p}{m} - \frac{\varepsilon_{rel}}{G_{-h}} \right) \right] \quad (22)$$

Compared to the non-pious $g^{h'}(\lambda)$, pious preferences impact sign $(g^{ph'}(\lambda))$ ambiguously.

As before, $g^{ph'}(\lambda) > (<) 0$ if numerator and denominator of the second factor in approximation (22) have opposite (the same) signs. The public good, a Church, is likely complementary to transfers, whence $f^{ph} > 0$, $\varepsilon_{rel} > 0$. Empiricists from Feldstein (1975) onwards find religious giving less price-elastic than non-religious giving, $(1 + \varepsilon_p^p + f_{rel}^{ph}) > 1 + \varepsilon_p^p > 1 + \varepsilon_p$. All else equal, religiosity makes $g^{ph'}(\lambda)$ more negative if $(1 + \varepsilon_p^p + f_{rel}^{ph}) (> 0)$ and $(1/f^{ph} - \lambda^{-1}(\varepsilon_m^p/m) - (\varepsilon_{rel}/G_{-h})) (> 0)$ are respectively larger and smaller than their non-pious counterparts in Equation (16).¹⁵

Standard comparative statics (see the Appendix) give Proposition 3 for sign $(g^{ph'}(\lambda))$:

Proposition 3. (i) $\text{sign}(g^{ph'}(\lambda))$ is a priori ambiguous; (ii) If $U(x, G, T)$ is additively separable, $-GU_{GG}/U_G \geq 1$ ensures $g^{ph'}(1) < 0$; (iii) If U is Cobb–Douglas, $U(x, G, T) = U[x^h, G, (1 - \lambda)G/\lambda] = \alpha_1 \log x^h + \alpha_2 \log G + \alpha_3 \log [1 + (1 - \lambda)G/\lambda]$, $\alpha_i \in (0, 1)$, $i = 1, 2, 3$, then $g^{ph'}(1) < 0$, $g^{ph'}(\lambda) < 0$, if $U(x, G, T)$ is log-Cobb–Douglas with no Inada condition re T .

The 1 in the third argument of $U[\cdot]$ stops transfers satisfying an Inada condition. Standard comparative statics again agrees with directly calculating the derivative of $g^{ph} = \left(f^{ph} \left[m^h + \frac{G_{-h}}{\lambda}, \left(\frac{1-\lambda}{\lambda} \right) G_{-h}, p_G^h \right] \right) / \lambda n^h$. Ultimately, $\text{sign}(g^{ph'})$ is resolvable empirically or by explicit preferences. for example, log-Cobb–Douglas U and budget constraint $x^h + (G/\lambda) = m^h + (G_{-h}/\lambda)$ give an h 's maxi-

¹⁵Other combinations are possible. For example, $f_{rel}^{ph} > 0$ might produce: (i) $(1 + \varepsilon_p^p + f_{rel}^{ph}) > 0$ if $(1 + \varepsilon_p^p) < 0$, and $[(1/f^{ph}) - (1/\lambda)(\varepsilon_m^p/m) - (\varepsilon_{rel}/G_{-h})] < 0$ if $[(1/f) - (1/\lambda)(\varepsilon_m/m)] < 0$ in the non-pious case; (ii) $(1 + \varepsilon_p^p + f_{rel}^{ph}) > 0$ if $(1 + \varepsilon_p) > 0$ and $[(1/f^{ph}) - (1/\lambda)(\varepsilon_m^p/m) - (\varepsilon_{rel}/G_{-h})] < 0$ if $[(1/f) - (1/\lambda)(\varepsilon_m/m)] > 0$ in the non-pious case. These could change $g^{h'}(\lambda) < 0$ to $g^{ph'}(\lambda) > 0$.

mand, $\alpha_1 \log (m^h + (G_{-h}/\lambda) - (G/\lambda)) + \alpha_2 \log G + \alpha_3 \log[1 + (1 - \lambda)G/\lambda]$. Its FOC yields a quadratic in G :

$$\left(m^h + \frac{G_{-h}}{\lambda}\right) \lambda^2 \alpha_2 + \left[(1 - \lambda) \lambda \left(m^h + \frac{G_{-h}}{\lambda}\right) (\alpha_2 + \alpha_3) - \lambda (\alpha_1 + \alpha_2)\right] G - (1 - \lambda) (\alpha_1 + \alpha_2 + \alpha_3) G^2 = 0 \quad (23)$$

This solves for G in terms of an h 's full income, $m^h + G_{-h}/\lambda$, parameters, α_i , and shares $(\lambda, 1 - \lambda)$. As $G = \lambda n^h g^{ph}$ and $G_{-h} = \lambda (n^h - 1) g^{ph}$ in Equation (23), Equation (24) defining g^{ph} implicitly in terms of λ gives an h 's Nash equilibrium donation and constrains the charity's maximisation:

$$\alpha_2 m^h + [(\alpha_2 + \alpha_3)(1 - \lambda)m^h n^h - (\alpha_2 + \alpha_1 n^h)] g^{ph} - (1 - \lambda)n^h (\alpha_2 + \alpha_3 + \alpha_1 n^h) g^{ph^2} = 0 \quad (24)$$

4.1 | The charity's maximisation with pious preferences

In a symmetric Nash equilibrium, if only the rich donate, social welfare is now

$$W^p \equiv n^h U[m^h - g^h, (1 - \lambda)n^h g^h, \lambda n^h g^h] + n^l U\left[m^l + \left(\frac{(1 - \lambda)}{n^l}\right) n^h g^h, (1 - \lambda)n^h g^h, \lambda n^h g^h\right] \quad (25)$$

To show $\lambda < 1$ is sub-optimal with non-pious preferences, W 's possible non-concavity in λ makes $\partial W / \partial \lambda|_{\lambda=1} > 0$ insufficient: $W(\lambda) > W(1)$ might hold at $\lambda < 1$. With pious preferences, for $\lambda < 1$ to be optimal for large n^h and/or n^l , $\partial W / \partial \lambda|_{\lambda=1} < 0$ suffices as some transfers dominate none. Let subscript T show derivatives w.r.t. T . Simplifying with an h -type's FOC, omitting functional arguments and superscript p in household utility,

$$\frac{\partial W^p}{\partial \lambda}|_{\lambda=1} = [n^h U_G^h + n^l U_G^l - n^h U_T^h - n^l U_T^l - U_x^l] n^h g^h + [(n^h - 1) U_G^h + n^l U_G^l] n^h g^{h'} \quad (26)$$

Comparing Equations (26) and (15) shows pious preferences makes transfers more socially worthwhile. A population effect deterring transfers no longer holds as their altruistic externality, $n^h U_T^h + n^l U_T^l$, increases with population, like the offsetting public good marginal benefit, $n^h U_G^h + n^l U_G^l$. Transfers are worthwhile if the utility from the first increment exceeds that from the marginal amount of public good were all donations used for the latter.¹⁶ Pairwise comparing terms in Equation (26) at $\lambda = 1$, $U_G[m^h - g^h(1), n^h g^h(1), 0] - U_T[m^h - g^h(1), n^h g^h(1), 0] \leq 0$ and $U_G[m^l, n^h g^h(1), 0] - U_T[m^l, n^h g^h(1), 0] \leq 0$ suffice for consumers to value transfers.

I examine this with log-Cobb–Douglas utility. By Proposition 3, $g^{ph'}(1) < 0$ then. So, $n^h g^{ph}(1) [n^h U_G^h + n^l U_G^l - n^h U_T^h - n^l U_T^l - U_x^l]|_{\lambda=1} \leq 0$ suffices for $\partial W / \partial \lambda|_{\lambda=1} < 0$, by Equation (26). Now, $g^{ph}(1) = [(\alpha_2 m^h) / (\alpha_2 + \alpha_1 n^h)]$ and $G^p(1) = [(\alpha_2 m^h n^h) / (\alpha_2 + \alpha_1 n^h)]$, as with non-pious preferences. Thus,

$$n^h g^{ph} [n^h U_G^h + n^l U_G^l - n^h U_T^h - n^l U_T^l - U_x^l]|_{\lambda=1} = n^h g^{ph}(1) \left[(n^h + n^l) \left(\frac{\alpha_2}{G^p} - \alpha_3 \right) - \left(\frac{\alpha_1}{m^l} \right) \right] \begin{cases} > \\ = \\ < \end{cases} 0 \text{ as}$$

¹⁶ W 's non-concavity could still make it increase in λ at $\lambda = 1$, yet W still attain its maximum at $\lambda < 1$.

$$\begin{aligned} & (n^h + n^l) \left[(\alpha_2 + \alpha_1 n^h) m^l - \alpha_3 n^h m^h m^l \right] - \alpha_1 n^h m^h \\ &= m^l \left\{ (n^h + n^l) \left[(\alpha_2 + \alpha_1 n^h) - \alpha_3 n^h \frac{m^h}{m^l} \right] - \alpha_1 n^h \frac{m^h}{m^l} \right\} \begin{cases} > \\ = \\ < \end{cases} 0 \end{aligned} \tag{27}$$

The expression in the last line is negative for plausible parameter and U.S. congregation sizes. A larger community does not make $\partial W / \partial \lambda|_{\lambda=1} < 0$ less likely now. The larger is inequality (i.e., m^h/m^l), and a need for transfers (because m^l is low), all else equal, the more negative is $-\alpha_1 n^h (m^h/m^l)$ and more likely is $\partial W / \partial \lambda|_{\lambda=1} < 0$. Next, as $\alpha_i \in (0, 1)$, $i = 1 - 3$, the size of $\alpha_1 n^h$ relative to $\alpha_3 n^h m^h$ mainly decides sign $[(\alpha_2 + \alpha_1 n^h) - \alpha_3 n^h m^h]$. This is more likely negative, so $\partial W / \partial \lambda|_{\lambda=1} < 0$ and transfers are more likely, the more the rich can afford them as m^h is large.

Table 1 exemplifies numerical optima with pious log-Cobb–Douglas preferences, showing $1 > \lambda^p > 0$ is plausible. The charity maximises W^p subject to Equation (24) from donors' optimisation. Throughout, $(n^h + n^l)$ is a realistic 180, $n^h/n^l = 5$, $m^h = 100$ and $m^l = 20$. Only parameters $\alpha_i \in (0, 1)$, $i = 1 - 3$, change. The exercise shows poverty-alleviating transfers can:

- (i)

greatly increase total welfare—in case A, by over 50% (from 482.22 to 730.91).
- (ii)

so increase giving, G hardly changes. In case A, with just over half total donations used for transfers, G is nearly 98% of its no-transfers level—0.647 06 cf. 0.662 25. The smallest ratio of G with transfers to that without is in B at nearly 90%.
- (iii)

significantly increase the poor's private consumption. The smallest increase occurs in C, at just over 4%; the largest is in A at over 16.5%.

TABLE 1 Illustrative simulations (via Maple) of giving with Pious log Cobb–Douglas preferences

$m^h = 100; m^l = 20; n^h = 150; n^l = 30$						
Case	Parameters	With transfers			Without transfers	
		Giving (g)	Welfare	l 's % Consumption rise	Giving (g)	Welfare
A	$\alpha_1 = \alpha_2 = \alpha_3 = 0.3$	$g = 1.3093$ $\lambda = 0.4942$	730.91	16.56	0.66225	482.22
B	$\alpha_1 = \alpha_2 = 0.3; \alpha_3 = 0.25$	$g = 1.2016$ $\lambda = 0.49504$	680.93	15.17	0.66225	482.22
C	$\alpha_1 = 0.4; \alpha_2 = \alpha_3 = 0.1$	$g = 0.32579$ $\lambda = 0.49363$	428.31	4.12	0.16639	370.07
D	$\alpha_1 = 0.4; \alpha_2 = 0.1; \alpha_3 = 0.2$	$g = 0.49098$ $\lambda = 0.32778$	440.82	8.25	0.16639	370.07
E	$\alpha_1 = 0.45; \alpha_2 = \alpha_3 = 0.225$	$g = 0.65574$ $\lambda = 0.49374$	668.26	8.3	0.33232	509.37
F	$\alpha_1 = 0.45; \alpha_2 = 0.225; \alpha_3 = 0.21$	$g = 0.63390$ $\lambda = 0.5073$	655.07	7.81	0.33232	509.37

Also:

- (iv) the relative weights of G and T in U decides λ . Equal or near-equal weighting gives λ close to 0.5. As T 's weight (α_3) rises relative to G 's (α_2), the share of giving used for transfers ($1 - \lambda$) increases. for example, in case D, when T 's weight is nearly twice G 's, nearly twice as much of donations go to transfers as to the public good.
- (v) All else equal, the total weight of G and T relative to x^h in U decides an h 's donation. The greater the x^h weighting, the less given.
- (vi) Without scope for transfers, giving and welfare are independent of the exponent on T in the utility function: compare giving and welfare in cases A and B, C and D, E and F.
- (vii) Simulated giving with pious preferences approximates that found in congregations, with over 1% of aggregate income donated and just under half used for poverty alleviation.

Additional numerical analysis (see the Appendix S2) studies giving and the division of its use between the pure public good and transfers as the proportion of the poor within a congregation of fixed size varies. I show that, in the log-Cobb–Douglas case again, as the proportion of the poor increases, the proportion of donations that the charity devotes to transfers increases, as does giving by each donor.

5 | OTHER PREFERENCES

5.1 | Warm-glow/joint-product preferences

In a pure warm-glow model, there is one public good and a number of private goods equal to 1 plus the number of agents in the model. The public good has quantity equal to the total donations of the agents to provide it from their endowment of the first (and common) private good. Each agent's donation to providing the public good gives them a warm glow, which is a private good for them only. With pious preferences, there is one private good (the endowment good) and two public goods (the total transfers, functioning as a public good, and the Church).

Would a utilitarian charity for many donors with joint-product/warm-glow preferences find it welfare-optimal to alleviate poverty? No. This is seen by applying my approach.

With warm-glow/joint-product preferences, $U(m^h - g^h, \lambda g^h + G_{-h}, g^h)$ is an arbitrary rich person's utility. I assume that the third argument (which generates warm glow to that person from giving) has no Inada condition: tastes force no-one to give.

Let subscript w on U denote its derivative re warm glow and superscript w identify magnitudes. I extend Assumption 4 (i) to Assumption 7 for plausibility and monotone comparative statics:

Assumption 7. (Supermodularity) $U_{ij} \geq 0$, $i \neq j$, $i, j = x, G, g$.

In stage-2's equilibrium now, an h person's FOC is

$$\begin{aligned} -U_x(m^h - g^h, \lambda g^h + G_{-h}, g^h) + \lambda U_G(m^h - g^h, \lambda g^h + G_{-h}, g^h) \\ + U_g(m^h - g^h, \lambda g^h + G_{-h}, g^h) = 0 \end{aligned} \quad (28)$$

A symmetric equilibrium with only rich donors gives a charity maximand

$$W^w = n^h U(m^h - g^h, \lambda n^h g^h, g^h) + n^l U[m^l + (1 - \lambda)n^h g^h/n^l, \lambda n^h g^h, 0]$$

I seek to show that the charity is unlikely to find transfers optimal in large congregations if agents have warm-glow/joint-product preferences. As in Section 3, I want $\partial W^w / \partial \lambda|_{\lambda \leq 1} > 0$ for large n^h and/or n^l . Differentiating W^w , using the h 's person's FOC,

$$\frac{\partial W^w}{\partial \lambda}|_{\lambda \leq 1} = n^h g^{wh} [n^h U_G^h + n^l U_G^l - U_x^l] + [(n^h - 1) \lambda U_G^h + \lambda n^l U_G^l + (1 - \lambda) U_x^l] n^h g^{wh'} \quad (29)$$

Equation (29) mirrors Equation (15) for non-pious preferences without warm glow. Proposition 4 (proven in the Appendix S1) shows $\partial W^w / \partial \lambda|_{\lambda \leq 1} > 0$ can occur in large communities.

Proposition 4. In Equation (29), (i) $[n^h U_G^h + n^l U_G^l - U_x^l] > 0$ if n^h and/or n^l are large; (ii) $\text{sign}(g^{wh'}(\lambda))$ is generally ambiguous. With $U(x, G, g)$ non-separable, $\text{sign}(g^{wh'}(\lambda))$ depends on which of warm glow and private consumption is more complementary with the public good, and on the elasticity of the marginal utility of G (i.e., $-GU_{GG}^h/U_G^h$). With $U(x, G, g)$ additively separable,

$$g^{wh'}(\lambda) \begin{pmatrix} > \\ = \\ < \end{pmatrix} 0 \text{ as } -GU_{GG}^h/U_G^h \begin{pmatrix} > \\ = \\ < \end{pmatrix} 1. \text{ With } U \text{ log-Cobb-Douglas, } g^{wh'}(\lambda) = 0.$$

Numerical analysis of the Cobb–Douglas case verifies the theory just given. Compared to the no warm glow model of Section 3, warm glow does not increase the desirability of transfers in large populations.¹⁷

5.2 | Brown (1997) Preferences

Brown's (1997) donors get an altruistic externality from giving to raise the poor's per capita income. She does not consider the poor's welfare explicitly, but I do so when resources used for transfers could instead be used for a pure public good. Suppose the poor's per capita income creates the altruistic externality from transfers. Brown's utility functions then are $U(x^i, G, x^l)$, $i = h, l$. I show that such preferences are unlikely to give enough incentive for transfers in typical congregations.

Index this case by superscript “B”. In an equilibrium with only h -types donating, the utilitarian charity now maximises welfare, W^B ,

$$W^B = n^h U \left(m^h - g^h, \lambda n^h g^h, m^l + \frac{(1 - \lambda)n^h g^h}{n^l} \right) + n^l U \left[m^l + \frac{(1 - \lambda)n^h g^h}{n^l}, \lambda n^h g^h, m^l + \frac{(1 - \lambda)n^h g^h}{n^l} \right] \quad (30)$$

¹⁷Appendix S1 considers the case of concave warm glow from multiple giving where donations are used for more than one purpose. In principle, the donor can get different amounts of warm glow from the different uses to which a donation is put. Proposition 5 there shows that, unsurprisingly, having multiple sources of warm glow increases the ambiguity regarding the effect of large numbers on the desirability of transfers.

Let subscripts s (“sharing”) show derivatives w.r.t. the poor’s per capita income. Using the h -type’s FOC in a stage-2 Nash equilibrium where only h -types donate,

$$\begin{aligned} \frac{\partial W^B}{\partial \lambda} \Big|_{\lambda \leq 1} = & n^h g^{Bh} \left[n^h U_G^h + n^l U_G^l - \left(\frac{n^h}{n^l} \right) U_s^h - U_s^l - U_x^l \right] \\ & + n^h g^{Bh'} \left[(n^h - 1) \lambda U_G^h + \lambda n^l U_G^l + (1 - \lambda) \left(\frac{n^h}{n^l} \right) U_s^h + (1 - \lambda) U_s^l + (1 - \lambda) U_x^l \right] \end{aligned} \quad (31)$$

Having the poor’s per capita income as an externality for all arguably fits the idea of being your brothers’ keeper better than having total transfers as the externality, as with pious preferences. But, as my Introduction notes, in a fixed congregation, total transfers, $(1 - \lambda)n^h g^h$, and the poor’s per capita transfers and consumption, $(1 - \lambda)n^h g^h / n^l$ and $m^l + [((1 - \lambda)n^h g^h) / n^l]$, increase together. So, Brown tastes are not better on that basis. More importantly, Equation (31) and numerical analysis show Brown tastes are less likely than pious tastes to make transfers improve welfare in big groups. This is seen intuitively by noting that, with Brown preferences, a transfer dollar split equally yields total altruistic externality of $n^l U_s^l (n^l)^{-1} = U_s^l$ and $n^h U_s^h (n^l)^{-1}$ for the poor and rich, respectively, cf. $n^l U_s^l$ and $n^h U_s^h$ with pious preferences. Used instead on the pure public good, that dollar gives benefits proportional to the rich and poor population sizes with Brown preferences, just as with pious tastes. Thus, the positive externality from transfers increases less than in proportion to population, and the opportunity cost increases proportionately, with Brown preferences. Conversely, both the positive externality from transfers and the opportunity cost of transfers increase proportionately with the population with pious preferences. So transfers are less likely with Brown preferences than with pious preferences as the population size increases.

I adopt a two-pronged approach to verifying this insight. First, I note that this large-number effect is captured by the $\left[n^h U_G^h + n^l U_G^l - \left(\frac{n^h}{n^l} \right) U_s^h - U_s^l - U_x^l \right]$ term on RHS Equation (31). I can show that this term is unambiguously positive with a large population size. This can be done most simply with additively separable utility. Let $U(x^i, G, x^l) \equiv U^x(x^i) + U^G(G) + U^s(x^l)$, $i = h, l$, with concave increasing U^k , $k = x, G, s$. Then, the Appendix S1 shows $n^h U_G^h + n^l U_G^l - \left(\frac{n^h}{n^l} \right) U_s^h - U_s^l - U_x^l > 0$ in Equation (31) for “large” n^h and/or n^l with Brown preferences, unlike its pious tastes equivalent. For the $g^{Bh'}(\lambda)$ term in Equation (31), $g^{Bh'}(1) < 0$ holds with log-Cobb–Douglas tastes, and generally if $-G^B U_{GG}^h / U_G^h \geq 1$, as with pious tastes. Second, the signs of the $n^h g^{Bh}$ and $n^h g^{Bh'}$ terms in Equation (31) are likely to differ for large n^h and/or n^l . Therefore, I conduct a numerical analysis with Brown preferences, similar to that underlying Table 1 with pious preferences, to see if feasible sub-game perfect charitable equilibria involving transfers with Brown preferences are likely with typically-sized congregations (See Appendix S2 where I consider specifically the counterparts to cases A and C in Table 1, but with Brown preferences.). This analysis shows that, with log-Cobb–Douglas preferences, while the charity finds it optimal to make transfers in very small congregations, this is not so with even moderately-sized congregations. This contrasts with the results for pious preferences, all else equal. I summarise this finding as:

Proposition 6. *A utilitarian charity is less likely to make transfers to the poor in a large society if consumers have Brown preferences than if they have pious preferences.*

Maintaining the log-Cobb–Douglas assumption for both pious and Brown preferences also enables me to show that, all else equal, the pious preferences case is likely to generate more giving and private

transfers than that with Brown preferences. To see this, let the Brown preferences for an h -person take the form

$$U(x^i, G, x^l) = \alpha \log(m^h - g^h) + \beta \log \lambda \left[(n^h - 1) \bar{g}^h + g^h \right] \\ + \delta \log \left(m^l + \frac{(1 - \lambda) \left((n^h - 1) \bar{g}^h + g^h \right)}{n^l} \right),$$

for parameters $\alpha, \beta, \delta \in (0, 1)$, when only the h -type gives and each conjectures that each of the others of their type makes a donation $\bar{g}^h$. The respective elasticity parameters α, β , and δ are the counterparts of α_1, α_2 , and α_3 with log-Cobb–Douglas pious preferences defined in Proposition 3. With Brown preferences, it is easily seen that, in a Nash equilibrium with only the h -types giving, the arbitrary h -person's FOC for an interior optimum with all h 's behaving identically is

$$\beta m^h m^l n^l + [(1 - \lambda) m^h n^h (\beta + \delta) - (\beta + \alpha n^h) m^l n^l] g^{Bh} - (1 - \lambda) n^h [\beta + \delta + \alpha n^h] g^{Bh^2} = 0 \quad (24')$$

Comparing Equations (24') and (24) enables us to state the following (proven in the [Appendix](#)):

Proposition 7. *Other things equal, if $m^l n^l > 1$, if the charity sets a given $\lambda \in (0, 1)$ and an h -person's problem is concave, more is donated if agents have pious preferences than if they have Brown preferences.*

In the numerical analysis underlying Table 1, all cases considered have $m^l n^l > 1$. So, in these cases, other things equal, if the charity sets a given $\lambda \in (0, 1)$, more is donated if agents have pious preferences than would be donated if they have Brown preferences.

Lastly, I consider numerically the effect on giving of varying the proportion of the poor in a fixed population with Brown preferences. It can be shown for log-Cobb–Douglas preferences (see Appendix S2 for details) that, when transfers occur, the charity uses a smaller fraction of what is given for transfers when the proportion of the poor increases and each donor gives less. This is diametrically opposite to the finding with pious preferences where, as the proportion of the poor in the congregation increases, a larger share of donations is used for transfers and each donor gives more. These results follow because, with both Brown and pious preferences, donors can be induced to give more by the charity increasing the share of donations that is used for transfers. However, with Brown preferences, the more poor there are to share the total transfers—hence the less the impact on their per capita income from any given amount of transfer—the less the charity's incentive to use donations for transfers, the less the share of donations that is used for that purpose and the less donors are induced to give.

6 | CONCLUSION

This paper shows that simultaneous charitable provision of private transfers and a separate public good that compete for resources can optimally occur with what I call “pious” preferences in a large population, yet this is not so with warm-glow and other preferences. In showing this, I identify a new large-number effect different from that usually associated with free-riding behaviour when charitable donations contribute to only a public good. “Pious preferences” are “prosocial” by internalising the amount given to the poor, reflecting the religious precept to love neighbours like yourself. They are in

the spirit of Becker as everyone internalising aggregate transfers to the poor makes this a public good. I show also that a charity bundling transfers and public good supply when people have pious tastes can greatly increase giving compared to a case of non-pious preferences. I think this explains the large amount of giving to religion and use of much of the receipts for the poor.

My analysis works precisely because aggregate private transfers to the poor function as a public good. When they do not, as with BBV, Andreoni warm-glow or Brown preferences, where aggregate private transfers are not analysed explicitly, the analysis does not work in the sense that the second large-number effect militating against transfers that I identify operates to prevent transfers being optimal in a large congregation.

Explanations for giving to religious charities differing superficially from mine exist. These must also stress utility from transfers to explain transfers. For example, churchgoers might give as they think their Church more trustworthy than other organizations. But this leaves using donations for transfers unexplained. Again, parishioners could feel closer to their own Church's poor members than to others'. But how do we model a “closer connection”, except as preference between types reflected in the utility function? For example, greater weight could be put on transfers to their own religion or race (cf. Alesina et al., 1999). As “11 o'clock Sunday morning continues to be the most segregated hour in America” (Hartford Institute, 2015), much religious giving allows donors to improve welfare only for those with whom they identify. People might also think that giving to the poor increases their chance of heaven. If this matters to them, they must have an in-life evaluation of post-life utility. This must increase in giving—for either the probability of getting to heaven to enjoy the post-life utility or/and the post-life utility itself increases with giving in life (as a good giver sits closer to the Deity). But, in all cases, this is reflected in the utility of transfers.

Several authors (e.g., Andreoni & Payne, 2013; Feldstein, 1975; List, 2011) suggest religious giving differs from other forms. It is less cyclical (List, 2011), less price-elastic (Feldstein, 1975), and a donor's piety decides the amount given (McKittrick et al., 2013). So, its motivation is not narrowly economic. Empathetic agents whose utility responds positively to total giving to the poor capture this. However, religious precepts might make worshippers feel moral constraints others do not—constraints not subsumed in utility functions, unlike with pious preferences. The constraints could reflect norms for giving—for example, tithing. Donors could then suffer a loss for how much their giving deviates from the “norm”, and balance disutility from giving (via lost consumption) with that loss. I will pursue this in future.

Lastly, I have ignored administration costs not directly benefiting charity targets. My G is not funding to maintain the charity's structure. Usually, consumers do not see administration costs as a public good, much less comparable to a Church building or pastor.

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CONFLICT OF INTEREST

The authors declared that they have no conflicts of interest to this work.

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APPENDIX

Proof of Proposition 1. In a symmetric Nash equilibrium for an arbitrary n^h with only the h type contributing at $\lambda \leq 1$, the optimal g^h satisfies FOCs $\lambda U_G [m^h - g^h(\lambda), \lambda n^h g^h(\lambda)] = U_x [m^h - g^h(\lambda), \lambda n^h g^h(\lambda)]$ and $\lambda U_G [m^l + \left(\frac{1-\lambda}{n^l}\right) n^h g^h(\lambda), \lambda n^h g^h(\lambda)] \leq \left(\frac{n^l + \lambda - 1}{n^l}\right) U_x [m^l + \left(\frac{1-\lambda}{n^l}\right) n^h g^h(\lambda), \lambda n^h g^h(\lambda)]$ (see inequality (17a) and Equation (17b)). The first FOC solves implicitly for g^h as a function of λ , m^h and n^h , independent of m^l and n^l , written as $g(\lambda)$ with dependence on m^h and n^h suppressed. In the expression for signing, consider $n^l U_G [m^l + ((1-\lambda)/n^l) n^h g^h(\lambda), \lambda n^h g^h(\lambda)] - U_x [m^l + ((1-\lambda)/n^l) n^h g^h(\lambda), \lambda n^h g^h(\lambda)]$. With $\lambda n^h g^h(\lambda)$ fixed for type 1, by concavity $U_x [m^l + ((1-\lambda)/n^l) n^h g^h(\lambda), \lambda n^h g^h(\lambda)]$ increases in n^l to a finite maximum $U_x [m^l, \lambda n^h g^h(\lambda)]$ when $n^l \rightarrow \infty$. As $((1-\lambda)/n^l) n^h g^h(\lambda)$ decreases in n^l , by supermodularity (Assumption 4(i)) $U_G [m^l + ((1-\lambda)/n^l) n^h g^h(\lambda), \lambda n^h g^h(\lambda)]$ decreases to a finite limit $U_G [m^l, \lambda n^h g^h(\lambda)]$. Hence, as n^l increases, $\left(U_x [m^l + \left(\frac{1-\lambda}{n^l}\right) n^h g^h(\lambda), \lambda n^h g^h(\lambda)] \right) / \left(U_G [m^l + \left(\frac{1-\lambda}{n^l}\right) n^h g^h(\lambda), \lambda n^h g^h(\lambda)] \right)$ attains a maximum, $(U_x [m^l, \lambda n^h g^h(\lambda)]) / (U_G [m^l, \lambda n^h g^h(\lambda)])$. The sum of the last two terms in the expression to be signed is non-negative if $n^l \geq U_x [m^l, \lambda n^h g^h(\lambda)] / U_G [m^l, \lambda n^h g^h(\lambda)]$. So, for $n^l \geq \hat{n}^l$, with $\hat{n}^l$ finite and defined by $\lim_{n \rightarrow \infty} \frac{U_x [m^l + \left(\frac{1-\lambda}{n^l}\right) n^h g^h(\lambda), \lambda n^h g^h(\lambda)]}{U_G [m^l + \left(\frac{1-\lambda}{n^l}\right) n^h g^h(\lambda), \lambda n^h g^h(\lambda)]} = \frac{U_x [m^l, \lambda n^h g^h(\lambda)]}{U_G [m^l, \lambda n^h g^h(\lambda)]} \equiv \hat{n}^l$, this sum is non-negative. Then the entire term for signing is positive, as its first term is. Also, $U_G [m^h - g^h(\lambda), \lambda n^h g^h(\lambda)] \geq U_G [m^l + \left(\frac{1-\lambda}{n^l}\right) n^h g^h(\lambda), \lambda n^h g^h(\lambda)]$.

$\lambda n^h g^h(\lambda)$ by $U_{GX} \geq 0$. So, I can instead combine the first and third terms in the expressions for signing to conclude that their sum is non-negative, thus the whole expression is positive if $n^h \geq n^l$. ■

Proof of Proposition 3. (i) Consider λ such that $g^{ph}(\lambda) > 0 = g^{pl}(\lambda)$. The FOC is

$$U_x [m^h - g^{ph}(\lambda), \lambda n^h g^{ph}(\lambda), (1 - \lambda)n^h g^{ph}(\lambda)] - \lambda U_G [m^h - g^{ph}(\lambda), \lambda n^h g^{ph}(\lambda), (1 - \lambda)n^h g^{ph}(\lambda)] \\ - (1 - \lambda)U_T [m^h - g^{ph}(\lambda), \lambda n^h g^{ph}(\lambda), (1 - \lambda)n^h g^{ph}(\lambda)] = 0 \quad (A1)$$

Differentiating Equation (A1) w.r.t. λ , omitting arguments and script h on utility terms,

$$g^{ph'}(\lambda) = \frac{n^h g^{ph} [U_{xG} - U_{xT} - (1 - 2\lambda)U_{GT} - \lambda U_{GG} + (1 - \lambda)U_{TT}] + U_T - U_G}{U_{xx} - \lambda(n^h + 1)U_{xG} - (1 - \lambda)(n^h + 1)U_{xT} + 2\lambda(1 - \lambda)n^h U_{GT} + \lambda^2 n^h U_{GG} + (1 - \lambda)^2 n^h U_{TT}} \\ \equiv \frac{\Gamma}{\Delta} \quad (A2)$$

By the second-order condition, $\Delta < 0$. But, Γ is ambiguously-signed, so $g^{ph'}(\lambda)$ is. At $\lambda = 1$,

$$g^{ph'}(1) = \frac{n^h g^{ph} [U_{xG} - U_{xT} + U_{GT} - U_{GG}] + U_T - U_G}{U_{xx} - (n^h + 1)U_{xG} + n^h U_{GG}} \equiv \Gamma_1 / \Delta_1 \quad (A3)$$

Again, $\Delta_1 < 0$ by concavity and supermodularity, but $g^{ph'}(1)$ is ambiguously-signed as Γ_1 is.

(ii) If $U(x, G, T)$ is additively separable, Equation (A2) becomes

$$g^{ph'}(\lambda) = \frac{n^h g^{ph} [-\lambda U_{GG} + (1 - \lambda)U_{TT}] + U_T - U_G}{U_{xx} + \lambda^2 n^h U_{GG} + (1 - \lambda)^2 n^h U_{TT}} \equiv \frac{\Gamma_2}{\Delta_2} \quad (A4)$$

So, as $n^h g^{ph} = G$ at $\lambda = 1$, $g^{ph'}(1) = [U_T - U_G - GU_{GG}] / [U_{xx} + n^h U_{GG}] < 0$ if $-GU_{GG}/U_G \geq 1$.

(iii) If $U(x, G, T)$ is log-Cobb–Douglas, $-GU_{GG}/U_G = 1$. From (ii), $g^{ph'}(1) < 0$. For $\lambda \in (0, 1)$, if $U(\cdot)$ does not satisfy Inada's condition re T , assume w.l.o.g. that the subutility for T is $\alpha_3 \ln(1 + T)$, $\alpha_3 \in (0, 1)$. Then $U_T = \alpha_3(1 + T)^{-1}$; $U_{TT} = -\alpha_3(1 + T)^{-2}$. In Γ_2 , $n^h g^{ph}(1 - \lambda)U_{TT} = TU_{TT} = -\alpha_3$

$$(1 + T)^{-2} \text{ and } n^h g^{ph}(1 - \lambda)U_{TT} + U_T = -\alpha_3 T(1 + T)^{-2} + \alpha_3(1 + T)^{-1} \begin{cases} > \\ = \\ < \end{cases} 0 \text{ as } 1 \begin{cases} > \\ = \\ < \end{cases} \frac{T}{1+T}. \text{ So}$$

$n^h g^{ph}(1 - \lambda)U_{TT} + U_T > 0$. Combining this with $\lambda n^h g^{ph} U_{GG} + U_G = 0$ in the log-Cobb–Douglas U , $\Gamma_2 > 0$, so $g^{ph'}(\lambda) < 0$ for $\lambda \leq 1$.

Proof of Proposition 7. Suppose α , β , and δ equal their counterparts α_1 , α_2 , and α_3 . Then Equations (24) and (24') only differ by the terms $m^l n^l [\beta m^h - (\beta + \alpha n^h) g^{Bh}]$ in the latter cf. $\beta m^h - (\beta + \alpha n^h) g^{ph}$ in the former. I confine attention to situations where, with pious preferences and redistribution, giving exceeds that without redistribution. In other words, $g^{ph} > \beta m^h / (\beta + \alpha n^h)$. So, at a given λ , m^h , m^l , n^h , and n^l , and at the same g ,

$$m^l n^l [\beta m^h - (\beta + \alpha n^h) g] \begin{cases} > \\ = \\ < \end{cases} [\beta m^h - (\beta + \alpha n^h) g] \text{ as } m^l n^l \begin{cases} < \\ = \\ > \end{cases} 1. \text{ So, Equations (24) and (24') imply}$$

$$\begin{aligned}
& \beta m^h m^l n^l + \left[(1 - \lambda) m^h n^h (\beta + \delta) - (\beta + \alpha n^h) m^l n^l \right] g - (1 - \lambda) n^h \left[\beta + \delta + \alpha n^h \right] g^2 \left\{ \begin{array}{l} > \\ = \\ < \end{array} \right\} \alpha_2 m^h + \\
& \left[(1 - \lambda) m^h n^h (\alpha_2 + \alpha_3) - (\alpha_2 + \alpha_1 n^h) \right] g - (1 - \lambda) n^h \left[\alpha_2 + \alpha_3 + \alpha_1 n^h \right] g^2 \\
& \text{as } m^l n^l \left\{ \begin{array}{l} < \\ = \\ > \end{array} \right\} 1. \text{ Now, if Equation (24) holds, if the arbitrary h-person's problem is concave,} \\
& g^{ph} \left\{ \begin{array}{l} > \\ = \\ < \end{array} \right\} g^{Bh} \text{ as } m^l n^l \left\{ \begin{array}{l} > \\ = \\ < \end{array} \right\} 1.
\end{aligned}$$